Homework 2: Linear Combinations, Span, and Homogeneous Systems

Application: 2D Movement in a Computer Game

A software development team is designing a 2D computer game. The position of a game character is represented by a vector $\boldsymbol{p} = \begin{bmatrix} x \\ y \end{bmatrix}$, where x represents horizontal position and y represents vertical position.

The game provides two basic movement commands:

$$\boldsymbol{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad \boldsymbol{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

A positive multiple of a vector represents movement in that direction, while a negative multiple represents movement in the opposite direction.

Part A — Linear Combination

The game needs to move a character from its current position to a target position represented by $w \ = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$.

1. Determine scalars a and b such that $a\boldsymbol{u} + b\boldsymbol{v} = \boldsymbol{w}$. Show all your working.
2. Write the resulting movement as a linear combination of $\boldsymbol{u}$ and $\boldsymbol{v}$. Explain in one or two sentences what the values of a and b mean in the context of the computer game.

Part B — Span

Suppose the game allows the programmer to use any real multiples of the two movement vectors.

3. Write the general movement that can be produced using $\boldsymbol{u}$ and $\boldsymbol{v}$: $a\boldsymbol{u} + b\boldsymbol{v}$. Simplify your answer.
4. Explain in your own words what $span\{\boldsymbol{u}, \boldsymbol{v}\}$ represents in this computer game. Your explanation should relate span to possible movements or reachable positions.
5. Determine whether the target position $\boldsymbol{q} = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$ can be generated using the two available movement commands. That is, determine whether there exist a, b such that $a\boldsymbol{u} + b\boldsymbol{v} = \boldsymbol{q}$. Show your work.

Part C — Homogeneous System

The developers now want to investigate whether different combinations of movement commands can produce zero net movement. They therefore consider $a\boldsymbol{u} + b\boldsymbol{v} = \boldsymbol{0}$.

6. Write this vector equation as a system of two linear equations in a and b.
7. Solve the homogeneous system.
8. What does your answer mean in the context of the game? In particular, explain what it means if there is only the trivial solution $a = b = 0$, or if there is a non-zero solution.

Part D — Matrix Representation

9. Write $a\boldsymbol{u} + b\boldsymbol{v} = \boldsymbol{w}$ in the form $A\boldsymbol{x} = \boldsymbol{w}$, where $\boldsymbol{x} = \begin{bmatrix} a \\ b \end{bmatrix}$. Clearly identify the matrix A.
10. Write the augmented matrix for the system and use elementary row operations to solve it.

Part E — Programming Task

Use Python to verify your answers. Your program should:

a) Define the vectors $\boldsymbol{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\boldsymbol{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

b) Define the target vector $\boldsymbol{w} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$.

c) Construct the matrix A.

d) Solve $A \begin{bmatrix} a \\ b \end{bmatrix} = \boldsymbol{w}$.

e) Display the values of a and b.

f) Verify the solution by calculating $a\boldsymbol{u} + b\boldsymbol{v}$.

g) Repeat the process for $\boldsymbol{q} = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$.

Challenge Question

Suppose the game designer replaces the second movement vector with $\boldsymbol{v} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$.

1. Can the game generate the target $\boldsymbol{w} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$?
2. Find the solutions of $a\boldsymbol{u} + b\boldsymbol{v} = \boldsymbol{0}$.
3. Explain what your results tell you about the available movement commands. Focus only on linear combinations, span, and homogeneous systems.
4. Determine whether the lines $\boldsymbol{x} = \boldsymbol{p} + s\boldsymbol{u}$ and $\boldsymbol{x} = \boldsymbol{q} + t\boldsymbol{v}$ intersect and, if they do, find their point of intersection, where

$$\boldsymbol{p} = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \boldsymbol{q} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \boldsymbol{u} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \boldsymbol{v} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$